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Advertising on the Web

CS246: Mining Massive Datasets
Jure Leskovec, Stanford University
Charilaos Kanatsoulis, Stanford University
<http://cs246.stanford.edu>



Announcements

- HW 4 and Colab 8 are due Thursday (3/5)
- Practice Final Recitation will be hosted on Wednesday 3/11, 4:30-6PM (details on Ed)
- HW/Lecture Review Section will be hosted on Saturday 3/14 4:00-5:30PM (details on Ed)

Actions on course feedback

- 2 review sessions (1 exam review session, 1 lecture/HW review session)
- Guest lecture from Anthropic to discuss the latest advancements of the field
- Focus more on the algorithmic details during class

Online Algorithms

- **Classic model of algorithms**

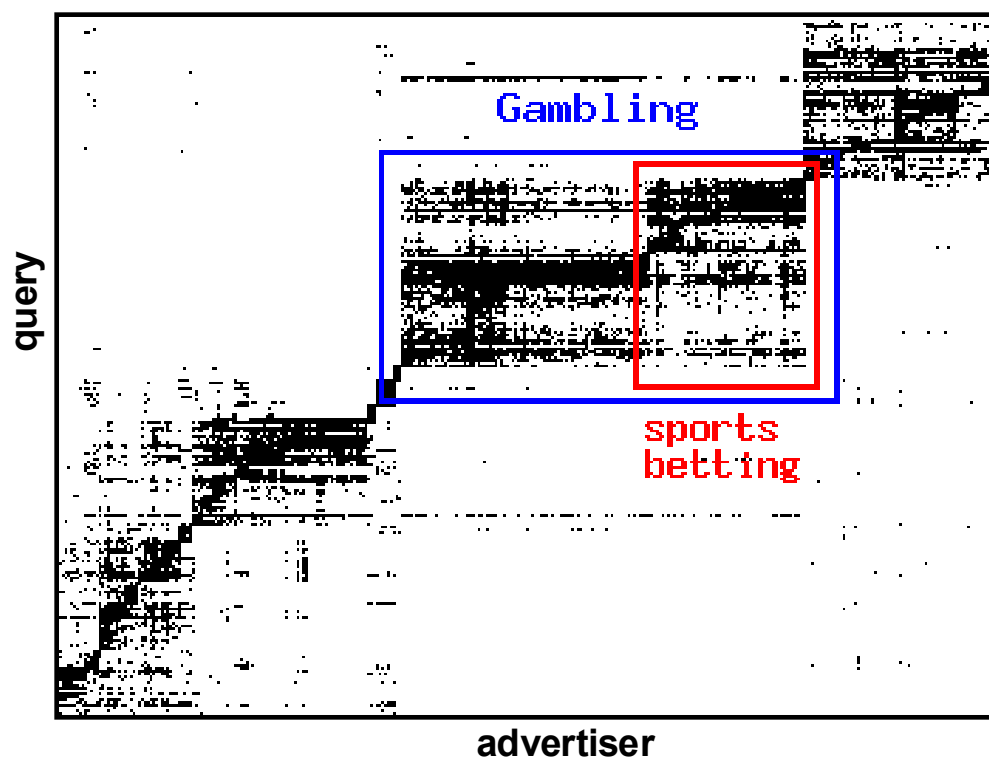
- You get to see the entire input, then compute some function of it
- In this context, “**offline** algorithm”

- **Online Algorithms**

- You get to see the input one piece at a time, and need to make irrevocable decisions along the way
- **Similar to the data stream model**

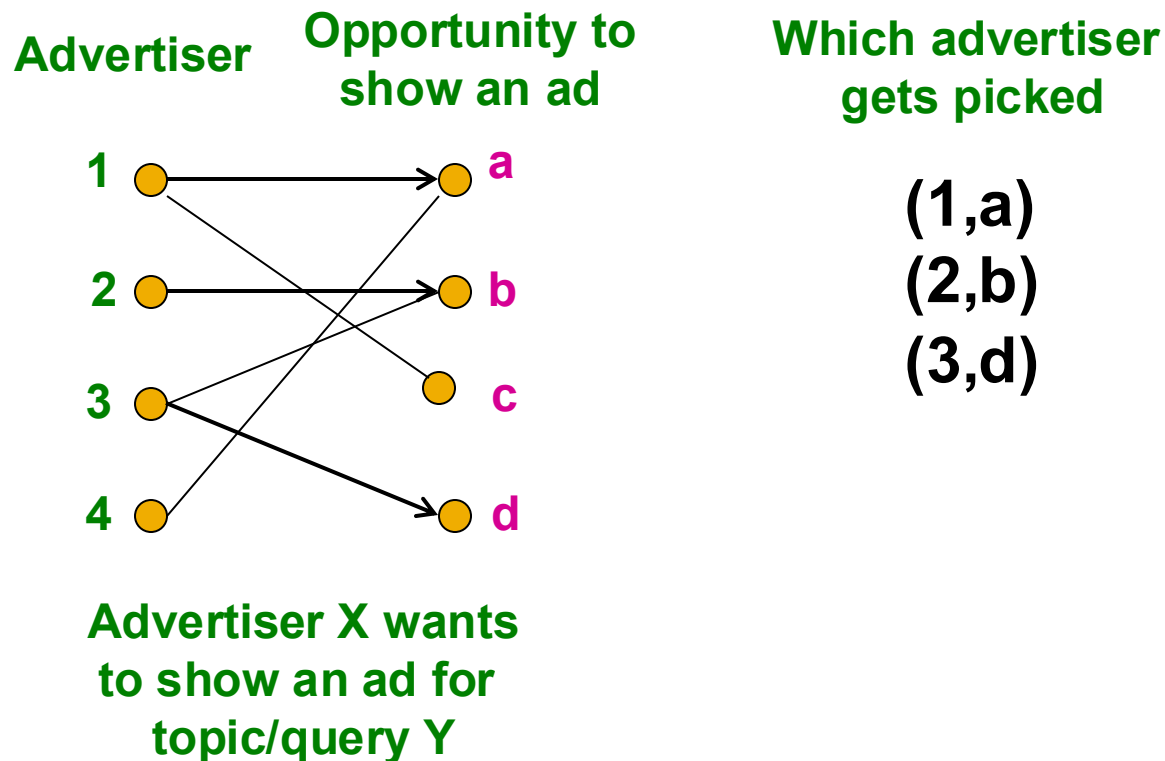
Sponsored Search: Ads

- Query-to-advertiser graph:



[Andersen, Lang: Communities from seed sets, 2006]

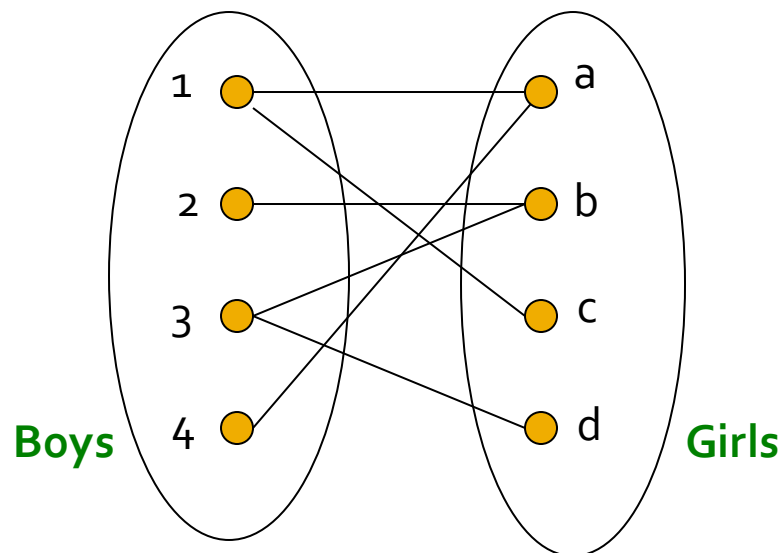
Graph Matching for Advertising



This is an online problem: We have to make decisions as queries/topics show up. We do not know what topics will show up in the future.

Online Bipartite Matching

Example: Bipartite Matching

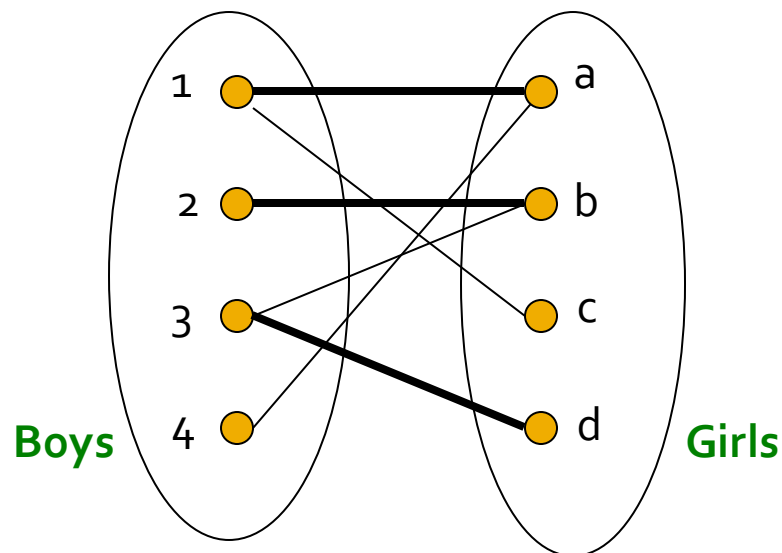


Nodes: Boys and Girls; Links: Preferences

Goal: Match boys to girls so that the most preferences are satisfied

Note: edges are only preferences with no weight or order.

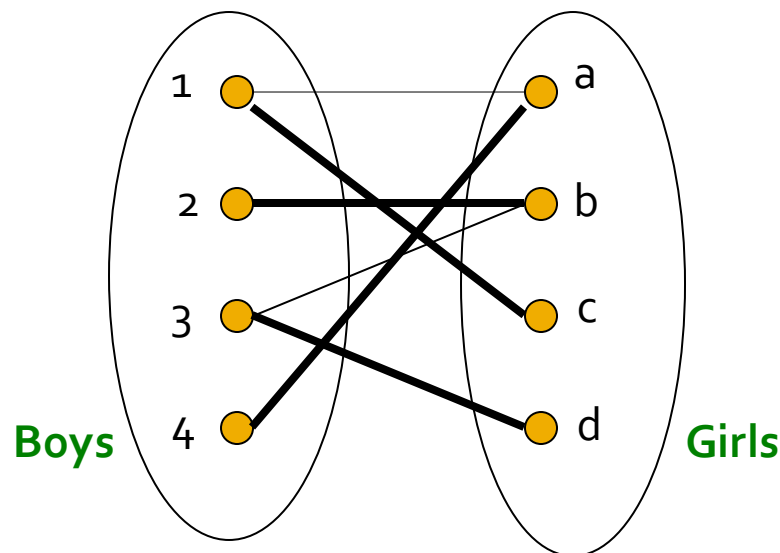
Example: Bipartite Matching



$M = \{(1,a), (2,b), (3,d)\}$ is a **matching**
Cardinality of matching = $|M| = 3$

Matching means that we are not using any vertex twice

Example: Bipartite Matching



$M = \{(1,c), (2,b), (3,d), (4,a)\}$ is a
perfect matching

Perfect matching ... all vertices of the graph are matched

Maximum matching ... matching that contains the largest possible number of matches

Matching Algorithm

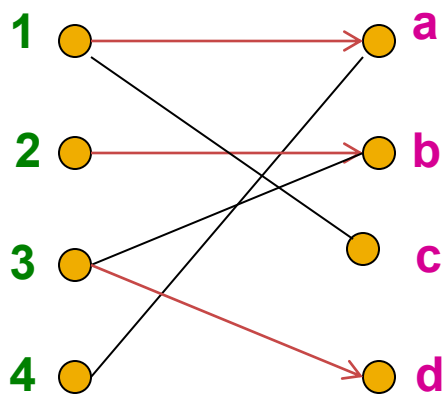
- **Problem:** Find a maximum matching for a given bipartite graph
 - A perfect one if it exists
- There is a polynomial-time offline algorithm based on augmenting paths (Hopcroft & Karp 1973, http://en.wikipedia.org/wiki/Hopcroft-Karp_algorithm)
- **But what if we do not know the entire graph upfront?**

Online Graph Matching Problem

- Initially, we are given the set boys
- In each round, one girl's choices are revealed
 - That is, the girl's edges are revealed
- At that time, we have to decide to either:
 - Pair the girl with a boy
 - Do not pair the girl with any boy
- Example of application:
Assigning tasks to servers

Note: Matching means that we are not using any girl or boy twice

Online Graph Matching: Example



(1,a)
(2,b)
(3,d)

Greedy Algorithm

- Greedy algorithm for the online graph matching problem:
 - Pair the new girl with **any** eligible boy
 - If there is none, do not pair the girl
- How good is the algorithm?

Competitive Ratio

- For input I , suppose greedy produces matching M_{greedy} while an optimal matching is M_{opt}

Competitive ratio =

$$\min_{\text{all possible inputs } I} (|M_{greedy}| / |M_{opt}|)$$

(what is greedy's worst performance over all possible inputs I)

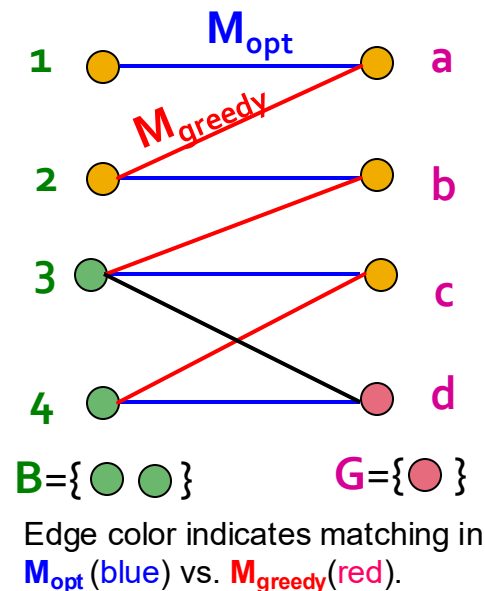
Analyzing the Greedy Algorithm

- Consider a case: $M_{greedy} \neq M_{opt}$
- Consider the set G of girls matched in M_{opt} but not in M_{greedy}

- (1) By definition of G :
$$|M_{opt}| \leq |M_{greedy}| + |G|$$

- (2) Define set B of boys linked to girls in G
 - Notice boys in B are already matched in M_{greedy} . Why?
 - If there would exist such non-matched (by M_{greedy}) boy adjacent to a non-matched girl then greedy would have matched them

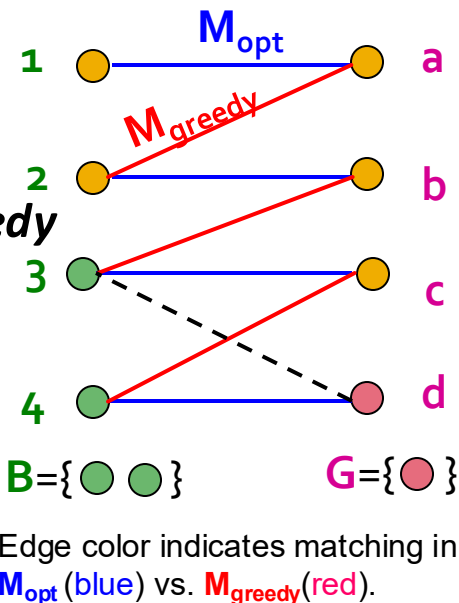
So: $|M_{greedy}| \geq |B|$



Analyzing the Greedy Algorithm

■ Summary so far:

- Girls G matched in M_{opt} but not in M_{greedy}
- Boys B adjacent to girls in G
- (1) $|M_{opt}| \leq |M_{greedy}| + |G|$
- (2) $|M_{greedy}| \geq |B|$



- Optimal matches all girls in G to (some) boys in B
 - (3) $|G| \leq |B|$
- Combining (2) and (3):
 - (4) $|G| \leq |B| \leq |M_{greedy}|$

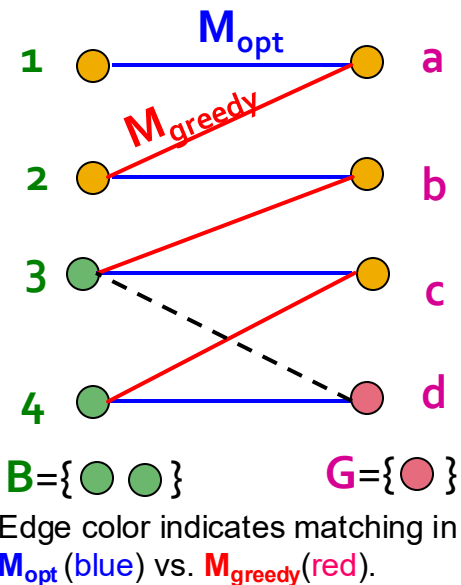
Analyzing the Greedy Algorithm

■ So we have:

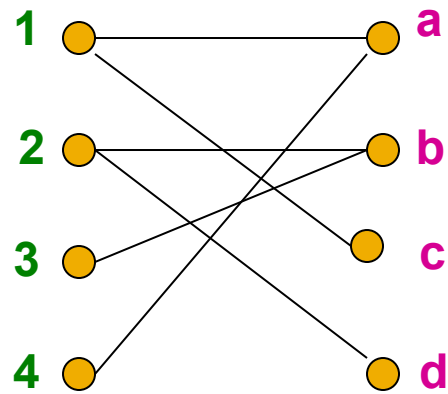
- (1) $|M_{opt}| \leq |M_{greedy}| + |G|$
- (4) $|G| \leq |B| \leq |M_{greedy}|$

■ Combining (1) and (4):

- Worst case is when $|G| = |B| = |M_{greedy}|$
- $|M_{opt}| \leq |M_{greedy}| + |M_{greedy}|$
- Then $|M_{greedy}| / |M_{opt}| \geq 1/2$



Worst-case Scenario



(1,a)
(2,b)

Web Advertising

History of Web Advertising

- **Banner ads (1995-2001)**

- Initial form of web advertising
- Popular websites charged \$X for every 1,000 “impressions” of the ad
 - Called “**CPM**” rate
(Cost per thousand impressions)
 - Modeled similar to TV, magazine ads
- From **untargeted** to **demographically targeted**
- **Low click-through rates**
 - Low ROI for advertisers



CPM...cost per mille
Mille...thousand in Latin

Performance-based Advertising

- Introduced by Overture around 2000
 - Advertisers **bid** on **search keywords**
 - When someone searches for that keyword, the **highest bidder's ad is shown**
 - Advertiser is charged only if the ad is clicked on
- Similar model adopted by Google with some changes around 2002
 - Called **Adwords**

Ads vs. Search Results

Web

Results 1 - 10 of about 2,230,000 for **geico**. (0.04 sec)

[GEICO](#) Car Insurance. Get an auto insurance quote and save today ...

GEICO auto insurance, online car insurance quote, motorcycle insurance quote, online insurance sales and service from a leading insurance company.

[www.geico.com/](#) - 21k - Sep 22, 2005 - [Cached](#) - [Similar pages](#)

[Auto Insurance](#) - [Buy Auto Insurance](#)

[Contact Us](#) - [Make a Payment](#)

[More results from www.geico.com »](#)

[Geico](#), Google Settle Trademark Dispute

The case was resolved out of court, so advertisers are still left without legal guidance on use of trademarks within ads or as keywords.

[www.clickz.com/news/article.php/3547356](#) - 44k - [Cached](#) - [Similar pages](#)

Google and [GEICO](#) settle AdWords dispute | The Register

Google and car insurance firm **GEICO** have settled a trade mark dispute over ... Car insurance firm **GEICO** sued both Google and Yahoo! subsidiary Overture in ...

[www.theregister.co.uk/2005/09/09/google_geico_settlement/](#) - 21k - [Cached](#) - [Similar pages](#)

[GEICO](#) v. Google

... involving a lawsuit filed by Government Employees Insurance Company (**GEICO**). **GEICO** has filed suit against two major Internet search engine operators, ...

[www.consumeraffairs.com/news04/geico_google.html](#) - 19k - [Cached](#) - [Similar pages](#)

Sponsored Links

[Great Car Insurance Rates](#)

[Simplify Buying Insurance at Safeco](#)

[See Your Rate with an Instant Quote](#)

[www.Safeco.com](#)

[Free Insurance Quotes](#)

[Fill out one simple form to get multiple quotes from local agents.](#)

[www.HometownQuotes.com](#)

[5 Free Quotes. 1 Form.](#)

[Get 5 Free Quotes In Minutes!](#)

[You Have Nothing To Lose. It's Free](#)

[sayyessoftware.com/Insurance](#)

[Missouri](#)

Web 2.0

- **Performance-based advertising works!**
 - Multi-billion-dollar industry
- **Interesting problem:**
Which ads to show for a given query?
 - (Today's lecture)
- **If I am an advertiser, which search terms should I bid on and how much should I bid?**
 - (Not focus of today's lecture)

AdWords Problem

- A stream of queries arrives at the search engine: $q_1, q_2, \dots$
- Several advertisers bid on each query
- When query q_i arrives, search engine must pick a subset of advertisers to show their ads
- **Goal:** Maximize search engine's revenues
 - **Simple solution:** Instead of raw bids, use the “expected revenue per click” (i.e., $\text{Bid} \times \text{CTR}$)
- **Clearly we need an online algorithm!**

The Adwords Innovation

Advertiser	Bid	CTR	Bid * CTR
A	\$1.00	1%	1 cent
B	\$0.75	2%	1.5 cents
C	\$0.50	2.5%	1.25 cents

Click through
rate

Expected
revenue

The Adwords Innovation

Advertiser	Bid	CTR	Bid * CTR
B	\$0.75	2%	1.5 cents
C	\$0.50	2.5%	1.25 cents
A	\$1.00	1%	1 cent

Instead of sorting advertisers by bid, sort by expected revenue

Limitations of Simple Algorithm

Instead of sorting advertisers by bid, sort by expected revenue

Advertiser	Bid	CTR	Bid * CTR
B	\$0.75	2%	1.5 cents
C	\$0.50	2.5%	1.25 cents
A	\$1.00	1%	1 cent

Challenges:

- CTR of an ad is unknown
- Advertisers have limited budgets and bid on multiple queries

Complications: Budget

- **Two complications:**
 - Budget
 - CTR of an ad is unknown

1) Budget: Each advertiser has a limited budget

- Search engine guarantees that the advertiser will not be charged more than their daily budget
 - The Ad Pacing problem:
 - Advertiser's ad works well on a popular query
 - Advertiser has a small budget and the budget gets exhausted quickly
 - Observing a click and recording it takes much longer and thus the advertiser gets a lot of free impressions/clicks

Complications: CTR

- **2) CTR (Click-Through Rate): Each ad-query pair has a different likelihood of being clicked**
 - **Advertiser 1** bids \$2 on query A, click probability = 0.1
 - **Advertiser 2** bids \$1 on query B, click probability = 0.5
- **CTR Prediction:**
 - ML models that predict the propensity of user clicking on a particular ad.
 - **Further complications: We will not cover:**
 - **1) CTR is position dependent:**
 - Ad #1 is clicked more than Ad #2

Complications: CTR

- **Complications we will cover:**

- **2) Exploration vs. exploitation**

Exploit: Should we keep showing an ad for which we have good estimates of click-through rate?

or

Explore: Shall we show a brand new ad to get a better sense of its click-through rate?

Online Algorithms

The BALANCE Algorithm

Adwords Problem

■ Given:

- 1. A set of bids by advertisers for search queries
- 2. A click-through rate for each advertiser-query pair
- 3. A budget for each advertiser (say for 1 month)
- 4. A limit on the number of ads to be displayed with each search query

■ Respond to each search query with a set of advertisers such that:

- 1. The size of the set is no larger than the limit on the number of ads per query
- 2. Each advertiser has bid on the search query
- 3. Each advertiser has enough budget left to pay for the ad if it is clicked upon

Greedy Algorithm

- **Our setting: Simplified environment**
 - There is **1** ad shown for each query
 - All advertisers have the same budget **B**
 - All ads are equally likely to be clicked
 - Bid value of each ad is the same (**$=\$1$**)
- **Simplest algorithm is greedy:**
 - For a query pick any advertiser who has bid **1** for that query
 - **Competitive ratio of greedy is $1/2$**

Bad Scenario for Greedy

- **Two advertisers A and B**
 - A bids on query x , B bids on x and y
 - Both have budgets of \$4
- **Query stream: $x x x x y y y y$**
 - Worst case greedy choice: $B B B B _ _ _ _$
 - Optimal: $A A A A B B B B$
 - Competitive ratio = $\frac{1}{2}$
- **This is the worst case!**
 - **Note:** Greedy algorithm is deterministic – it always resolves draws in the same way

BALANCE Algorithm [MSVV]

- **BALANCE** Algorithm by Mehta, Saberi, Vazirani, and Vazirani
 - For each query, pick the advertiser with the largest unspent budget
 - Break ties arbitrarily (**but in a deterministic way**)

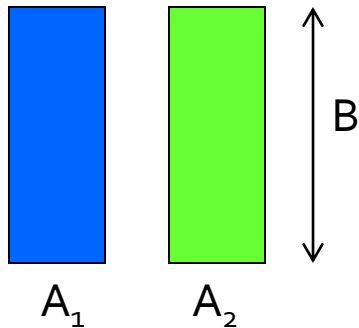
Example: BALANCE

- **Two advertisers A and B**
 - A bids on query x , B bids on x and y
 - Both have budgets of \$4
- **Query stream:** $x x x x y y y y$
- **BALANCE choice:** A B A B B B _ _
 - Optimal: A A A A B B B B
- **In general:** For **BALANCE** on 2 advertisers
Competitive ratio = $\frac{3}{4}$

Analyzing BALANCE

- **Consider simple case (w.l.o.g.):**
 - 2 advertisers, A_1 and A_2 , each with budget B (≥ 1)
 - Optimal solution exhausts both advertisers' budgets
- **BALANCE must exhaust at least one budget:**
 - **If not, we can allocate more queries**
 - Whenever BALANCE makes a mistake (both advertisers bid on the query), advertiser's unspent budget only decreases
 - Since optimal exhausts both budgets, one will for sure get exhausted
 - Assume BALANCE exhausts A_2 's budget, but allocates x queries fewer than the optimal
 - **So revenue of BALANCE = $2B - x$** (where OPT is $2B$)
 - **Let's work out what x is!**

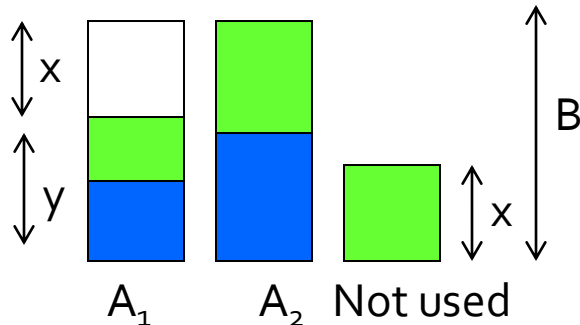
Analyzing Balance



■ Queries allocated to A_1 in optimal solution

■ Queries allocated to A_2 in optimal solution

Opt revenue = $2B$



Balance revenue = $2B - x = B + y$

We claim $y \geq x$ (next slide).

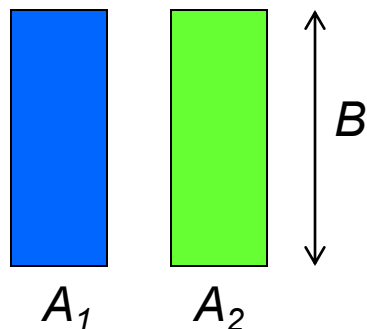
Balance revenue is minimum for $x=y=B/2$.

Minimum Balance revenue = $3B/2$.

Competitive Ratio = $3/4$.

Balance allocation

Analyzing BALANCE: What's x ?

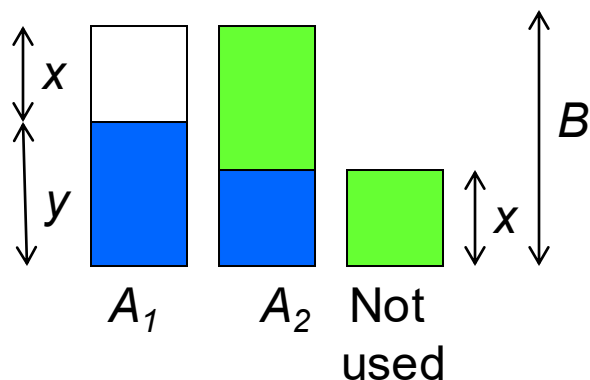


- Queries allocated to A_1 in the optimal solution
- Queries allocated to A_2 in the optimal solution

Optimal revenue = $2B$

Assume Balance gives revenue = $2B - x = B + y$

Assume we exhausted A_2 's budget



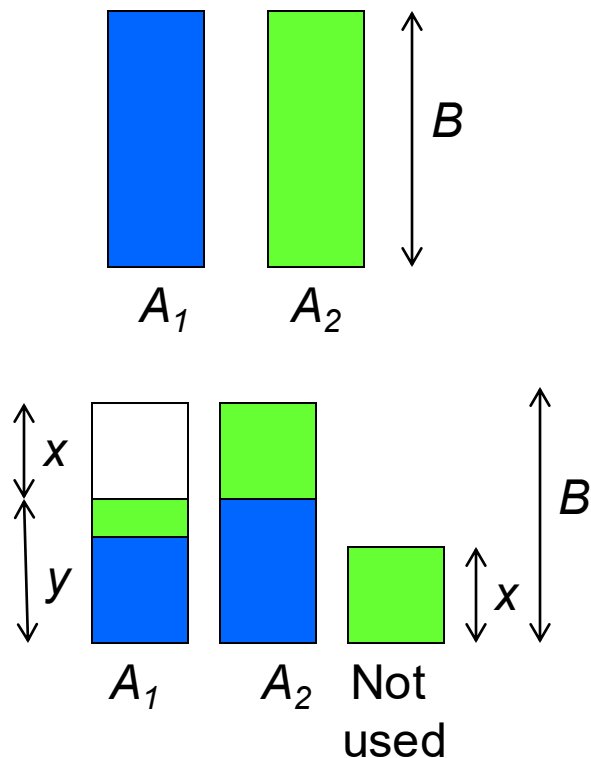
Notice: Unassigned (not used) queries should be assigned to A_2 (since if we could assign them to A_1 we would since we still have the budget)

Goal: Show we have $y \geq B/2$

Case 1) BALANCE assigns $\geq B/2$ blue queries to A_1 .

Then trivially, $y \geq B/2$

Analyzing BALANCE: What's x ?



■ Queries allocated to A_1 in the optimal solution

■ Queries allocated to A_2 in the optimal solution

Optimal revenue = $2B$

Assume Balance gives revenue = $2B - x = B + y$

Assume we exhausted A_2 's budget

Unassigned queries should be assigned to A_2
(if we could assign to A_1 we would since we still have the budget)

Goal: Show we have $y \geq B/2$

Balance revenue is minimum for $x = y = B/2$

Minimum Balance revenue = $3B/2$

Competitive Ratio: $BAL/OPT = 3/4$

Case 2) BALANCE assigns $> B/2$ blue queries to A_2 .

Consider the last blue query assigned to A_2 .

At that time, A_2 's unspent budget must have been at least as big as A_1 's.

That means at least as many queries have been assigned to A_1 as to A_2 .

At this point, we have already assigned at least $B/2$ queries to A_2 .

So, $x \leq B/2$ and $x + y = B$ then $y > B/2$

BALANCE: General Result

- In the general case, worst competitive ratio of BALANCE is $1 - 1/e = \text{approx. } 0.63$
 - $e = 2.7182$
 - Interestingly, no online algorithm has a better competitive ratio!
- Let's see the worst case example that gives this ratio

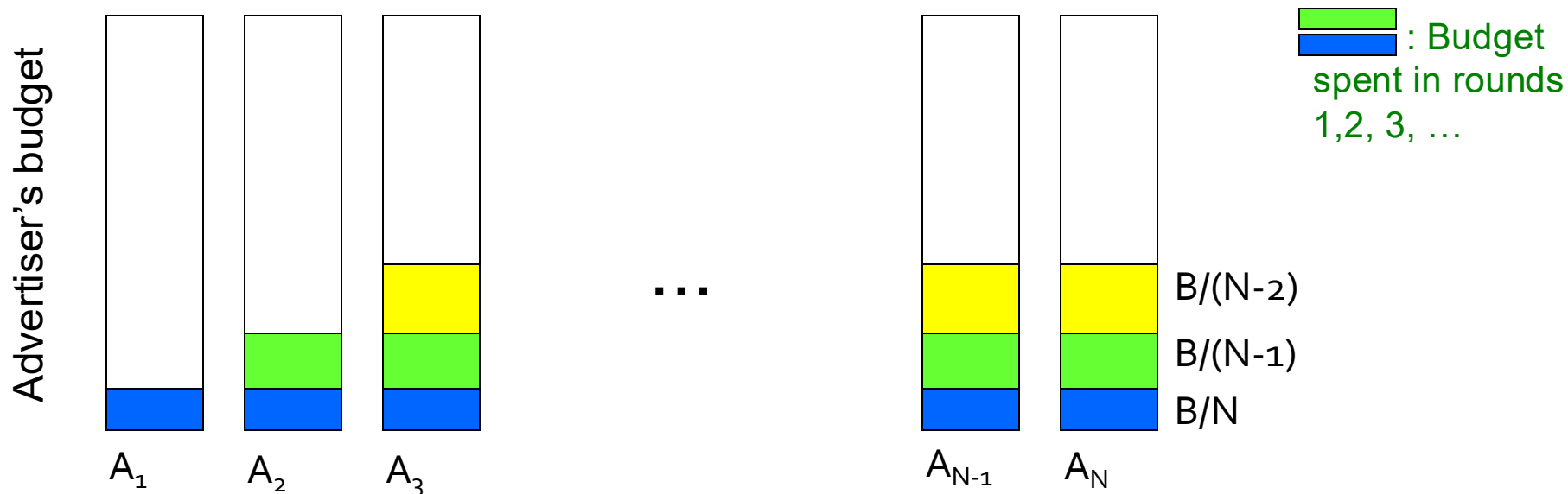
Worst case for BALANCE

- **N advertisers:** $A_1, A_2, \dots, A_N$
 - Each with budget $B > N$
- **Queries:**
 - $N \cdot B$ queries appear in N rounds of B queries each
- **Bidding:**
 - Round 1 queries: bidders $A_1, A_2, \dots, A_N$
 - Round 2 queries: bidders $A_2, A_3, \dots, A_N$
 - Round i queries: bidders $A_i, \dots, A_N$
- **Optimum allocation:**

Allocate all round i queries to A_i

 - Optimum revenue $N \cdot B$

BALANCE Allocation

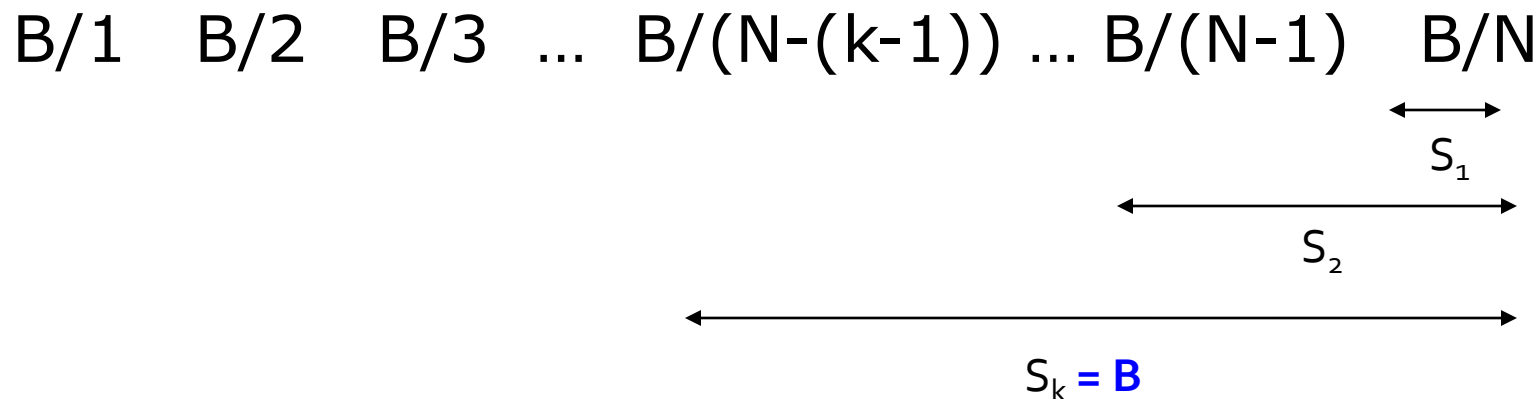


BALANCE assigns each of the queries in round 1 to **N** advertisers. After **k** rounds, **sum of allocations** S_k to each of advertisers

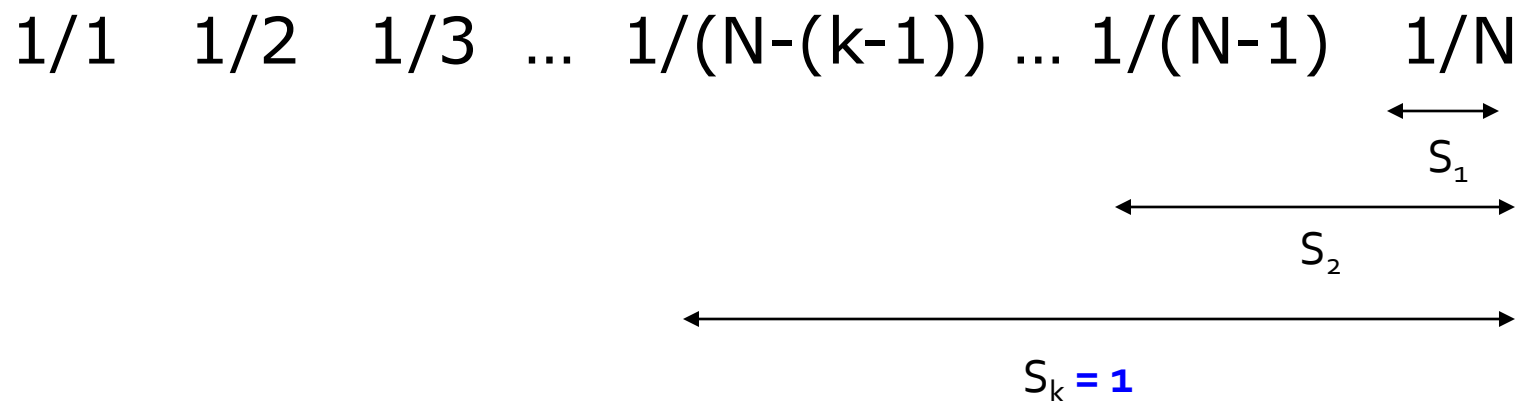
$$A_k, \dots, A_N \text{ is } S_k = S_{k+1} = \dots = S_N = \sum_{i=1}^k \frac{B}{N-(i-1)}$$

If we find the smallest k such that $S_k \geq B$, then after k rounds we cannot allocate any queries to any advertiser

BALANCE: Analysis



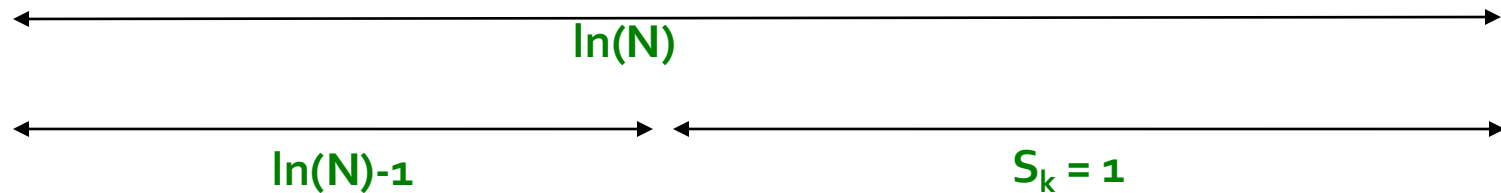
Can divide everything by B:



BALANCE: Analysis

- **Fact:** $H_n = \sum_{i=1}^n 1/i \approx \ln(n)$ for large n
 - Result due to Euler

$1/1 \quad 1/2 \quad 1/3 \quad \dots \quad 1/(N-(k-1)) \quad \dots \quad 1/(N-1) \quad 1/N$



- $S_k = 1$ implies: $H_{N-k} = \ln(N) - 1 = \ln\left(\frac{N}{e}\right)$
 - We also know: $H_{N-k} = \ln(N-k)$
 - So: $N - k = \frac{N}{e}$
 - Then: $k = N\left(1 - \frac{1}{e}\right)$
- N terms sum to $\ln(N)$.
Last k terms sum to 1.
First $N-k$ terms sum to $\ln(N-k)$ but also to $\ln(N)-1$*

BALANCE: Analysis

- So after the first $k=N(1-1/e)$ rounds, we cannot allocate a query to any advertiser
- Revenue = $B \cdot N (1-1/e)$
- Competitive ratio = $1-1/e$
- Note: So far we assumed:
 - All advertisers have the same budget B
 - All advertisers bid 1 for the ad
 - (but each advertiser can bid on any subset of ads)

General Version of the Problem

- **Arbitrary bids and arbitrary budgets!**
- Consider we have 1 query q , advertiser i
 - Bid = x_i
 - Budget = b_i
- **In a general setting BALANCE can be terrible**
 - Consider two advertisers A_1 and A_2
 - $A_1: x_1 = 1, b_1 = 110$
 - $A_2: x_2 = 10, b_2 = 100$
 - Consider we see **10** instances of q
 - BALANCE always selects A_1 and earns **10**
 - Optimal earns **100**

Generalized BALANCE

- **Arbitrary bids:** consider query q , bidder i
 - Bid = x_i
 - Budget = b_i
 - Amount spent so far = m_i
 - Fraction of budget left over $f_i = 1 - m_i/b_i$
 - Define $\psi_i(q) = x_i(1 - e^{-f_i})$
- Allocate query q to bidder i with largest value of $\psi_i(q)$
- **Same competitive ratio $(1 - 1/e) = 0.63$**